

May 15

Sketch of proof of Hamilton's 1982 Theorem.

Let (M^3, g_0) is closed & $\text{Ric}(g_0) > 0$.

$$\text{RF:} \quad \partial_t g = -2 \text{Ric}_g(t)$$
$$g(0) = g_0$$

Control the geometry of $g(t)$

Constant sectional curvature $\overset{n=3}{\iff} \text{Ric} = \frac{1}{3} R g$.
contracted second Bianchi identity.

$$\text{div}(\text{Ric}) = \frac{1}{2} \nabla R = \frac{1}{2} dR.$$

i.e. $g^{ik} \nabla_i R_{jk} = \frac{1}{2} \nabla_k R$.

$$\text{Ric} = \frac{1}{3} R g \implies R = \text{const.}$$

$$\frac{1}{2} dR = \text{div}(\text{Ric}) = \text{div}\left(\frac{1}{3} R g\right) = \frac{1}{3} dR \implies dR = 0.$$

$$\boxed{\partial_t R_{ij} = \underset{\text{smoothing}}{\Delta R_{ij}} + \underset{\text{driving}}{Q_{ij}}}$$

$$R_{ij} \geq 0$$

Hamilton's tensor maximum principle.

Since Q_{ij} has the property:

$$\text{If } R_{ij} V^i = 0, \text{ then } Q_{ij} V^i V^j \geq 0.$$

If $R_{ij} \geq 0$ initially, then $R_{ij} \geq 0$ for all time.

$\text{Ric} \geq \delta g$, $\delta > 0$. is a fixed amount of positivity

Consider the inequality

(*) $\boxed{\text{Ric} \geq \delta R g}$, $\delta > 0$.

(M^3, g_0) closed, $\text{Ric}(g_0) > 0 \Rightarrow R(g_0) > 0$
 $\Rightarrow \exists \delta > 0$ such that (*) is true.

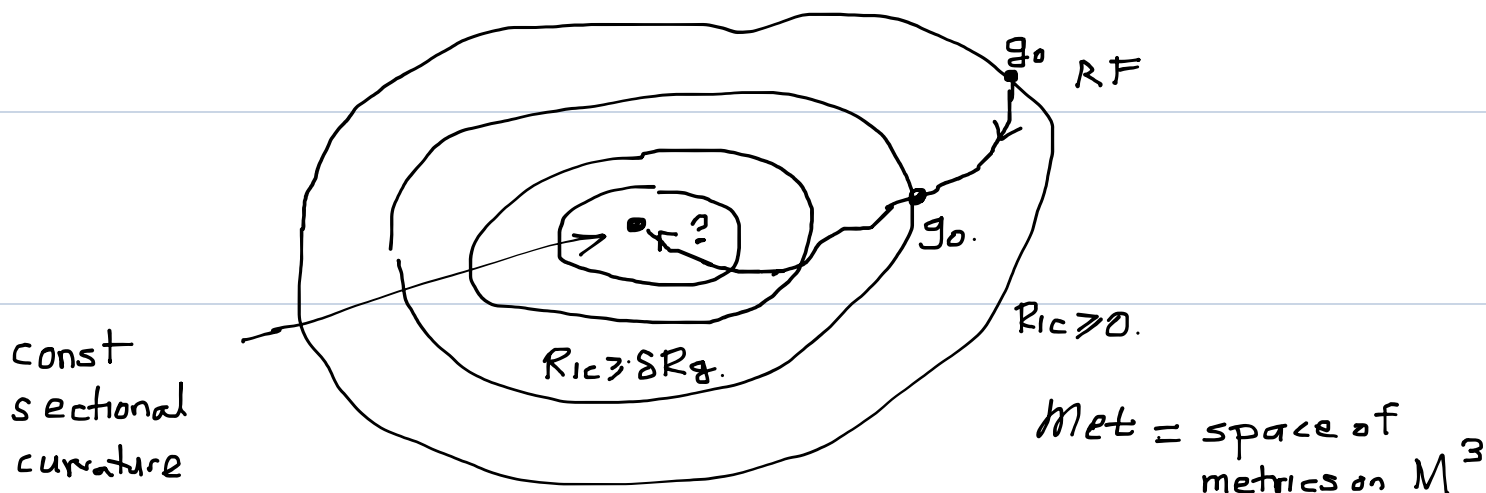
$$P := \text{Ric} - \delta R g.$$

To preserve (*). "Ricci pinching" is equivalent to showing $P \geq 0$ is preserved.

(**) $\frac{\partial}{\partial t} P_{ij} = \Delta P_{ij} + S_{ij}$.

Exercise: (**) satisfies "null-eigenvector condition". $\square$

Schematic diagram.



Remark: $\text{Ric} \geq \delta R g \Rightarrow \delta \leq \frac{1}{3}$.

In fact $Ric \geq \frac{1}{3}Rg \Rightarrow Ric = \frac{1}{3}Rg$.

$$Ric = \frac{1}{3}Rg \iff Ric - \frac{1}{3}Rg = 0.$$

$$\iff |Ric - \frac{1}{3}Rg|^2 = 0.$$

↗ a scalar.

$$\text{Let } F = F(g) := |Ric - \frac{1}{3}Rg|^2$$

$$F(cg) = c^{-2} F(g).$$

shrinking almost round sphere



$$g(t) = c(t)g, \quad c(t) \rightarrow 0 \text{ as } t \rightarrow T.$$

Suppose geom. doesn't improve.

What would happen to.

$$F(g(t))? \quad \text{expect } F \nearrow \infty.$$

Define

$$\begin{aligned} G = G(g) &= \frac{|Ric - \frac{1}{3}Rg|^2}{R^2} \\ &= \frac{|Ric|^2}{R^2} - \frac{1}{3} \end{aligned}$$

This is scale-invariant.

world

We like this to go to 0, i.e.,

we want

$$\frac{|Ric - \frac{1}{3}Rg|^2}{R^2} \rightarrow 0.$$

Showing $\square \rightarrow 0$ is harder than showing $\square \leq C$.

$$E := \frac{|Ric - \frac{1}{3}Rg|^2}{R^{2-\epsilon}} \quad \text{where } \epsilon > 0.$$

$$E \text{ scales like } \frac{1}{R^{-\epsilon}} = R^\epsilon$$

 as $t \rightarrow T$, expect $R \nearrow \infty$
so expect $R^\epsilon \nearrow \infty$ since $\epsilon > 0$.

Theorem. If (M^3, g_0) is a closed 3-manifold with positive Ricci curvature, then $\exists \epsilon > 0$

and $C = C(g_0, \epsilon)$ such that

$$E = \frac{|Ric - \frac{1}{3}Rg|^2}{R^{2-\epsilon}} \leq C \quad \text{on } M^3 \times [0, T).$$

$$\partial_t E = \Delta E + \frac{2(1-\epsilon)}{R} \nabla R \cdot \nabla E$$

$$- \frac{2}{R^{4-\epsilon}} |R \nabla Ric - \nabla R \otimes Ric|^2$$

$$|R \nabla_i R_{jk} - \nabla_i R R_{jk}|^2$$

$$- \frac{\epsilon(1-\epsilon)}{R^{4-\epsilon}} |Ric - \frac{1}{3}Rg|^2 |\nabla R|^2$$

$$+ \frac{2}{R^{3-\epsilon}} \left(\underbrace{\epsilon |Ric|^2 |Ric - \frac{1}{3}Rg|^2}_{\text{red box}} - Q \right)$$

where

$$Q := 2|Ric|^4 + R^4 - 5R^2 |Ric|^2 + 4R \operatorname{trace}(Ric^3).$$

Lemma (M^3, g) Riem.

If $R > 0$ & $\text{Ric} \geq \delta R g$, then.

$$Q \geq 2\delta^2 |\text{Ric}|^2 / |\text{Ric} - \frac{1}{3}Rg|^2$$

This an algebraic calculation.

Cor. If $\varepsilon \leq 2\delta^2$, then.

$$\frac{\partial}{\partial t} E \leq \Delta E + \chi(t) \cdot \nabla E.$$

So by the max. prin. $E \leq C$ on $M^3 \times [0, T)$.

How to finish off the proof?

'Normalized' RF :

$$(\#) \quad \partial_t g = -2\text{Ric} + \frac{2}{3}r g$$

where $r = r(t)$ = average scalar curvature at time t .

$$= \frac{\int_M R_{g(t)} dV_{g(t)}}{\int_M dV_{g(t)}}.$$

Exercise Under $(\#)$, $\text{Volume}(g(t)) = \text{Volume}(g_0)$.

Hamilton should :

• solution to $(\#)$ exists for all $t \in [0, \infty)$

• $g(t) \rightarrow$ Riem. metric, denoted by g_∞ .
as $t \rightarrow \infty$

$C^\infty \swarrow$

- g_∞ has constant positive sectional curvature,

Convergence is pointwise and derivatives of all orders.

$$|g_{ij}(t) - (g_\infty)_{ij}| \rightarrow 0 \text{ as } t \rightarrow \infty$$

$$\& \quad \left| \partial_I^k g_{ij}(t) - \partial_I^k (g_\infty)_{ij} \right| \rightarrow 0 \text{ as } t \rightarrow \infty \\ \forall k.$$

actually exponentially fast.

We not going to prove what Hamilton proved.

We will prove something weaker, namely:

$\exists$ g_∞ on M^3 with const. positive sectional curvature

Talk about convergence from a broader perspective.

(CG). Cheeger - Colding convergence of g_i ,

where $g_i = R_i g(t_i)$,

where $g(t)$ is a solution to RF

on $[0, T)$ and $t_i \rightarrow T$.

Want $g_i \xrightarrow{CG} \text{some } g_\infty$ on M^3 .

What is CG convergence? : just

C^∞ convergence in the usual pointwise sense

after we take into account diffeomorphisms.

CG compactness theorem. (pointed version).

Consider $g(t_i)$ and choose

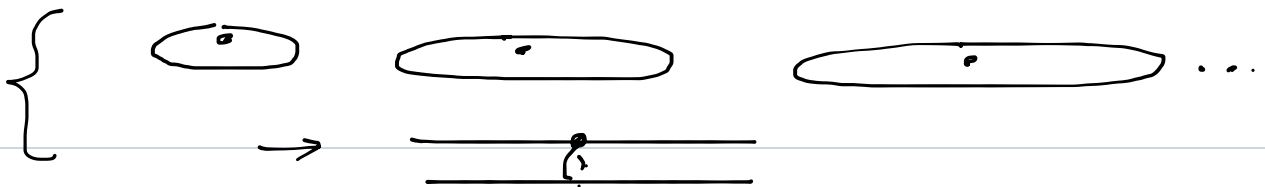
$x_i \in M^3$ such that

$$R_i := R(x_i, t_i) = \max_{x \in M^3} R(x, t_i).$$

So $\tilde{g}_i = R_i g(t_i)$ has scalar curvature

1 at x_i & scalar curvature ≤ 1 on M^3 .

$$(M^3, \tilde{g}_i, x_i) \xrightarrow{\text{CG}} (M_\infty^3, \tilde{g}_\infty, x_\infty).$$



$\tilde{g}_\infty$ has const. positive sectional curvature.

& complete.

$\Rightarrow M_\infty^3$ is compact. $\xrightarrow{\text{CG conv.}} M_\infty^3$ diffeo to M^3 .
